MIL-C-51358A(EA)
14 February 1978
SUPERSEDING
MIL-C-51358(MU)
28 August 1972

MILITARY SPECIFICATION

CANISTERS, SMOKE, HC 155MM

PROJECTILE, M1 and M2; COMPONENTS FOR

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the components for two types of canisters.

1.2 Classification. The canister components shall be furnished in the following types as specified (see 6.2):

Type M1 - Canister, Smoke, HC, 155MM Shell, Components For (TDPL15-11-121)
Type M2 - Canister, Smoke, HC, 155MM Shell, Components For (TDPL15-11-122)

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 1320

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-A-2550 - Ammunition General Specification For.
MIL-P-16232 - Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals).
MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat Sealable.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1167 - Ammunition Data Card.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL15-11-121 - Canister, Smoke, HC, 155MM Shell, M1; Components For.
TDPL15-11-122 - Canister, Smoke, HC, 155MM Shell, M2, Components For.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL15-11-121(M1) and TDPL15-11-122(M2) and on the subsidiary drawings shall conform to the specifications listed thereon or to the characteristics set forth on the drawings.

3.1.2 Components. All components of the canisters shall conform to the specifications and drawings listed on TDPL15-11-121 and TDPL15-11-122 and subsidiary drawings.

3.2 Assembly. The body and tube shall be in accordance with Drawings C15-11-39(M1), C15-11-47(M2), D15-11-40(M1), and D15-11-48(M2) and shall be assembled in accordance with the body assembly Drawings D15-11-38(M1) and C15-11-46(M2). Surface treatment shall be in accordance with applicable drawings and specifications.

3.3 Welds. The body assembly shall not contain more than three separated projections when tested as specified in 4.4.4.1.

3.4 Salt spray. The body assemblies and closing disks shall meet the requirements of MIL-P-16232 when tested as specified in 4.4.4.2.

3.5 Preproduction. Prior to the start of regular production, a preproduction lot shall be produced in accordance with this specification for examination and test (see 4.3).

3.6 Workmanship. The components shall be free from rust, contamination (foreign matter, oil, or grease), and damage such as dents, cracks, or deformation which could affect their intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Additional contractor's responsibility. Drawings for contractor designed inspection equipment (other than standard measuring equipment) shall be forwarded to the design agency for approval prior to use for

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production (see 6.2). For preproduction and regular production, the canister metal components manufacturer shall be responsible for the minimum mechanical property tests required by drawing C15-11-39 and C15-11-47. Tests shall be performed by a Government approved laboratory. Minimum mechanical property tests shall be found satisfactory prior to production. Provisions of MIL-A-2250 apply. Ammunition data cards prepared in accordance with MIL-STD-1167 shall be forwarded with each lot of components.

4.1.3 Government responsibility. Unless otherwise specified, the Government will be responsible for preproduction inspection. Preproduction samples (4.3.2) shall be forwarded to the design agency or laboratory designated by the contracting officer (see 6.2).

4.1.4 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 of which specific inspection has not been provided in this specification have been satisfied. Specifically laboratory test data and tool control dimensional data shall be provided to the contracting officer as proof of compliance of the material specified for the components.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 500 sets of components of the type specified (see 6.2) shall be manufactured using the same methods, materials, equipment, processes, and cycle time as will be used during regular production. Components consist of body, tube, body assembly, closure disk, zinc sleeve and packing disk. Samples as required by 4.3.2 shall be forwarded to the design agency or Government designated laboratory for inspection and evaluation prior to proceeding with regular production (see 6.2). Provisions of MIL-A-2550 shall apply.

4.3.2 Sampling.

4.3.2.1 For examination. Components sufficient to assemble 50 canisters of the type specified (see 6.2) shall be taken at random from the preproduction lot. Five samples each of the specified body, formed tube and tube flat development, shall be selected and checked against their respective drawings.

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4.3.2.2 For destructive tests.

(a) Body assembly and closing disk. Ten body assemblies of each type specified (see 6.2) and closing disks shall be taken at random from the preproduction lot.

(b) Body. Five bodies of each type specified (see 6.2) shall be taken at random from the preproduction lot for testing (see 4.1.2).

(c) Body assembly and closing disk after varnish. Five body assemblies of each type specified (see 6.2) with varnish applied per Dwg D15-11-38 and C15-11-46 and closing disks, shall be taken at random from the preproduction lot.

4.3.3 Inspection procedure.

4.3.3.1 For examination. The sample components shall be examined in accordance with the classification of defects (4.4.3.3) and MIL-STD-105. The individual bodies, and tubes in flat and formed configurations shall be examined for conformance to their respective drawings. The acceptance number is zero.

4.3.3.2 For destructive tests.

(a) Body assemblies. Five body assemblies of each type specified (see 6.2) shall be tested in accordance with 4.4.4.1. No body assembly shall contain more than three separated projections.

(b) Body. Five bodies shall be tested in accordance with the applicable drawings. The acceptance number is zero.

(c) Body assembly and closing disk. Five body assemblies of each type specified (see 6.2) and closing disk shall be tested in accordance with 4.4.4.2. Assemblies shall meet the requirements of MIL-P-16232 for type Z, class 3 for 2 hours.

(d) Body assembly and closing disk after varnish. Five body assemblies of each type specified (see 6.2) with varnish applied per Dwg D15-11-38 and D15-11-46 and closing disk, shall be tested in accordance with 4.4.4.2. The varnished areas shall meet the requirements of MIL-P-16232, for type Z, class 2 for 48 hours.

4.3.4 Acceptance/rejection criteria. The preproduction samples shall comply with 4.3.3 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. A lot shall consist of like components of one type or like subassemblies of one type, one interfix number, produced by one manufacturer, from the same materials and under essentially the same manufacturing conditions. No one lot of any component shall contain steel from more than one heat of steel. Provisions of MIL-A-2550 shall apply.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests.

(a) Body assembly and closing disk. Ten body assemblies of each type specified (see 6.2) (Drawing D15-11-38 and D15-11-46) and closing disks, shall be taken at random from each lot.

(b) Body. Five bodies of each type specified (see 6.2) (Drawing C15-11-39 and C15-11-47) shall be taken at random from production at increments not to exceed 50,000 bodies and when the manufacturing process changes or a new heat of steel is used. Testing shall be as specified in 4.1.2.

(c) Body assembly and closing disk after varnish. Five body assemblies of each type specified (see 6.2), with varnish applied per Dwg D15-11-38 and C15-11-46 and closing disk; shall be taken at random from each lot.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample components shall be examined in accordance with the classification of defects (4.4.3.3) and MIL-STD-105.

4.4.3.2 For destructive tests.

(a) Body assemblies. Five body assemblies of the type specified (see 6.2) shall be tested in accordance with 4.4.4.1. The acceptance number is zero.

(b) Body. Five bodies of the type specified (see 6.2) shall be tested in accordance with the applicable drawings. The acceptance number is zero.

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(c) Body assembly and closing disk. Five body assemblies of each type specified (see 6.2) and closing disk shall be tested in accordance with 4.4.4.2. Assemblies shall meet the requirements of MIL-P-16232, for type Z, class 3 for 2 hours.

(d) Body assembly and closing disk after varnish. Five body assemblies of each type specified (see 6.2), with varnish applied per Dwg D15-11-38 and C15-11-46 and closing disk, shall be tested in accordance with 4.4.4.2. The varnished area of the body assembly shall meet the requirements of MIL-P-16232, for type Z, class 2 for 48 hours.

4.4.3.3 Classification of defects. (See pages 8 thru 12).

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Welds.

(a) Secure assembly in suitable fixture which allows bottom tube weld area free movement.

(b) Place a 24 inch rigid hollow bar snugly over perforated tube (bar shall have a 1/8-inch maximum wall thickness and shall show no evidence of bending during test). Move bar laterally, touching diametrically opposite walls and return to center. one time only.

(c) Remove bar and remove assembly from fixture.

(d) Examine projection welds for metal separation, cracks or other failures.

4.4.4.2 Salt spray test. The body assembly and closing disk of the type specified (see 6.2) shall be subjected to a salt spray test in accordance with the requirements of MIL-P-16232.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing, level B.

5.1.1 Body assembly, metal (see 6.2). Twenty-four empty metal containers of the same type shall be packaged together in two layers of 12 canisters each. Each layer shall be arranged in four rows of three canisters per row. Inter-locking fiberboard partitions shall be used to separate the canisters from each other. In addition, full size separator pads shall be placed on the bottom of the canisters, between the two layers and on top of the canisters. The partitions and the separator pads shall be constructed of material conforming to grade V3c of PPP-F-320. This

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 5		DRAWING NUMBER
4.4.3.3(a)	Canisters, Smoke, HC, 155MM Shell, M1 and M2; Components For				D15-11-38 (M1) and C15-11-46(M2)Body Assembly
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
101	Perpendicularity incorrect		0.40%	3.1	Gage
102	Position incorrect		0.40%	3.1	Gage
103	Formed radius incorrect		0.40%	3.1	Gage
104	Tube OD incorrect		0.40%	3.1	Gage
105	Tube flange protrusion incorrect		0.40%	3.1	Scale
106	Tube hole missing		0.40%	3.1	Visual
107	Tube and flange butt joint space excessive after assembly		0.40%	3.1	Feeler gage or standard measuring wire
108	Body length incorrect		0.40%	3.1	Gage
109	Body OD incorrect		0.40%	3.1	Gage
110	Body thickness incorrect		0.40%	3.1	Standard measuring instruments (SMI)
111	Coating missing or incorrect		0.40%	3.1	Visual
112	Tube length from outside bottom of body to bottom of slot incorrect		0.40%	3.1	SMI/Gage
113	Flash tube will not accept plug gage		0.40%	3.1	Gage
114	Flash tube will not accept ring gage		0.40%	3.1	Gage
115	Workmanship (3.6)		0.40%	3.6	Visual

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PARAGRAPH	TITLE				DRAWING NUMBER
4.4.3.3(b)	Canisters, Smoke, HC, 155MM Shell, M1 and M2; Components For		SHEET 2 OF 5		C15-11-37, Disk, Closing
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
101	Outside diameter incorrect		0.40%	3.1	SMI
102	Hole diameter incorrect		0.40%	3.1	SMI
103	Hole position to outside diameter		0.40%	3.1	SMI
104	Coating incorrect		0.40%	3.1	Visual
105	Edge radius incorrect		0.40%	3.1	SMI
106	Thickness incorrect		0.40%	3.1	SMI
107	Workmanship (3.6)		0.40%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

MIL-C-51358A(EA)

PARAGRAPH	TITLE	SHEET 3 OF 5	DRAWING NUMBER
4.4.3.3(c)	Canisters, Smoke, HC, 155MM Shell, M1 and M2; Components For		B15-11-124 Disk, Packing

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
*101	Diameter incorrect		0.40%	3.1	Tool control
*102	Hole diameter incorrect		0.40%	3.1	Tool control
103	Thickness incorrect		0.40%	3.1	Certification

*Tool Control:

Tooling to be checked prior to and upon completion of production or any time tooling is modified or changed. Tooling must also be verified after each 40 hours of operation. Tooling must be correct at each check.

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PARAGRAPH	TITLE	SHEET 4 OF 5	DRAWING NUMBER
4.4.3.3(d)	Canisters, Smoke, HC, 155MM Shell, M1 and M2; Components For		C15-11-35(M1) and C15-11-43(M2) Sleeve, Zinc

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY TDPL PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
101	Thickness incorrect		0.40%	3.1	Certification
102	Length incorrect		0.40%	3.1	SMI
*103	Seam incorrect		0.40%	3.1	Tool control
104	Staking missing		0.40%	3.1	Visual
*105	Stake diameter incorrect		0.40%	3.1	Tool control
*106	Inside diameter incorrect		0.40%	3.1	Tool control
*107	Spotwelds incorrect (Alternate design)		0.40%	3.1	Visual/SMI

*Tool Control:

Tooling to be checked prior to and upon completion of production or any time tooling is modified or changed. Tooling must also be verified after each 40 hours of operation. Tooling must be correct at each check.

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MIL-C-51358A(EA)

PARAGRAPH	TITLE		SHEET 5 of 5		DRAWING NUMBER
4.4.3.3(e)	Canisters, Smoke, HC, 155MM Shell, M1 and M2; Components For				N.A.
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
101	Packaging materials incorrect		0.40%	5.1	Visual and certification
102	Heat sealed bag improperly closed or damaged		0.40%	5.1	Visual
103	Packaging component missing or damaged		0.40%	5.1	Visual

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assembly shall be encased in a heat sealed protective bag constructed of material conforming to type 1 of MIL-B-22191. The bagged items shall then be overpacked in a fiberboard box conforming to grade V3c, style FOL of PPP-B-636. The box, bag, and all interior dunnage shall not be larger than required to provide a snug fit when all of the package is in place. Closure of the fiberboard box shall be in accordance with the requirements of method V of the appendix to PPP-B-636.

5.1.2 Zinc sleeve. Zinc sleeves shall be packed in a manner to prevent damage or disfigurement of placing them in a shipping container utilizing interlocking partitions in which each sleeve shall occupy an individual cell. The quantity of sleeves that shall be packed together in a shipping container shall be commensurate with sound economic and efficient packing practices and shall assure safe delivery to a first receiving activity in an undamaged condition.

5.1.3 Disk, packing. Disks shall be packed together in a manner to prevent physical damage or disfigurement by placing them in a shipping container which shall provide sound economic and efficient packaging and provide safe delivery to a first receiving activity in an undamaged condition.

5.1.4 Disk, closing. Two-hundred disks of the same type shall be packed together, 20 stacks of ten disks per stack. The 20 stacks shall be arranged in five rows with four stacks per row. The stacks shall be separated from each other by interlocking partitions. Fiberboard separator pads constructed of material conforming to grade V3c of PPP-F-320 shall be placed on top and on the bottom of the configuration. This assembly shall then be placed within a heat sealed protective bag constructed of material conforming to type 1 of MIL-B-22191. The bagged items shall then be overpacked in a fiberboard box conforming to grade V3c, style FOL of PPP-B-636. The box, bag, and all interior dunnage shall not be larger then required to provide a snug fit when all of the package is in place. Closure of the fiberboard box shall be in accordance with method V of the appendix to PPP-B-636.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the domestic marking requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. These components are intended for M1 and M2 canisters to be used in the shell, Smoke, HC, BE, 155MM, M116E2.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of canister (M1 or M2) required.
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government inspection, test, and evaluation after award of contract.
(2) Name and address of Government facility conducting the preproduction inspection and address for sending results of minimum mechanical property tests.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.
(4) Name and address of Government facility approving surface treatemnt process.

(d) Design agency where drawings for contractor designed inspection equipment should be forwarded.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1320-A404

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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